

Muhammad Ibrahim

+971 (55) 892 8217 | mi2171@nyu.edu | [linkedin.com/in/ibrahim2411](https://www.linkedin.com/in/ibrahim2411) | No Sponsorship Needed | HPI Visa | UK Work Authorisation

EDUCATION

New York University

Bachelor of Science in General Engineering, Mechanical and Civil Focus

- **Major GPA - 3.60 / 4.00**, UK (2:1).
- **Major Coursework** - Solid Mechanics, Engineering Dynamics, Fluid Mechanics, Heat Transport, Composite Materials.

Abu Dhabi, UAE

May 2026

New York University

Study Away, Immigration and Theatre

- Analysed migration in French theatre casting and the economics of integration; **Term GPA - 3.70 / 4.00**.

Paris, France

Jan 2024

New York University June Term

Study Away, Architecture and Climate Impact

- Examined how architectural decisions shape climate outcomes and purpose-led design; **Term GPA - 4.00 / 4.00**.

Valletta, Malta

Jun 2023

EXPERIENCE

Center for Stability, Instability and Turbulence (SITE)

Graduate Intern, New York University

- Ran a 2D free surface study solo, **60 trials** across **4 angles of attack**, to build the dataset the model was trained on.
- Grounded the predictive model in a dimensional analysis of surface height against velocity, depth and body radius.
- Trained a **machine learning model in Python and MATLAB** predicting free surface profiles beyond the tested range.

New York, USA

May 2026 - Jul 2026

Advanced Manufacturing Workshop

Workshop Technician, New York University

- Cut print failure rates from **50% to 10%** across over **70 print hours** by diagnosing root causes and refining parameters.
- Manufactured polymer and composite parts, reinforcing them with bidirectional carbon fibre, fibreglass and chopped strand.

Abu Dhabi, UAE

Sep 2025 - May 2026

Center for Stability, Instability and Turbulence (SITE)

Undergraduate Research Assistant, New York University

- Tested over **130 fluid instability** cases in a 2-person team, probing for linearity between laminar and turbulent regimes.
- Developed water-tunnel models that reliably triggered **boundary layer separation**, quantified by flow visualisation.

Abu Dhabi, UAE

Apr 2023 - May 2026

Energy and Propulsion Lab

Undergraduate Research Assistant, New York University

- Solved **Computational Fluid Dynamics** problems in Converge and OpenFOAM on HPC clusters.
- Simulated airflow through vertical growing towers for **AeroFarms**, returning a **28%** ventilation gain at upper levels.
- Presented Urban Air Mobility propulsion conclusions to a **25-member Hyundai NGV panel** in South Korea.

Abu Dhabi, UAE

Nov 2022 - Nov 2023

PROJECTS

Fixed-Wing UAV for Search and Rescue Payload Deployment, *Personal Project*

Jun 2026 - Present

- Owned the airframe, structure and materials on a **1130 mm** wingspan 3D printed UAV, built with one collaborator.
- Reduced airframe mass from **2500 to 1500 g** via lightweight PLA, infill tuning, internal geometry and carbon tubing.
- Designed a release mechanism to drop a multirotor payload in flight, delivered as a completed prototype.

Advanced Anti-Roll Tank System (ART3S), *Mubadala Investment and Abu Dhabi Maritime Academy*

Oct 2024 - Sep 2025

- Engineered SolidWorks models of **200 to 800 cubic metre** anti-heeling tanks for full-scale ferry and car-carrier vessels.
- Modelled vessel roll decay in **Simulink**, cutting decay time **33%** with passive tanks and **62%** with active control.

Savonius Turbine for Traffic Wind Energy Harvesting, *Capstone Design*

Sep 2024 - May 2025

- Optimised a vertical-axis wind turbine in **ANSYS Fluent** with a **45-degree** helical twist and serrated leading edges.
- Raised energy recovery from **56%** of the theoretical maximum to **80%** by adding the serrations.
- Measured **3 to 15 W** output at **1 to 5 m/s** wind from a 1:3 scale carbon fibre FDM prototype.

Compressor-Less Multi-Stage TEC Refrigerator, *New York University*

Mar 2025 - May 2025

- Replaced compressor and refrigerant with **5 stacked thermoelectric modules**, cooling **27 to 10 degrees** in **24 minutes**.
- Tuned a closed-loop **PID controller** on Arduino, holding the **10 degree** setpoint within the **±0.5 degree** sensor accuracy.

Formula Student SAE Drivetrain Design, *New York University*

Sep 2024 - May 2025

- Specified the rear half shafts driving the wheels on a rear-engined Formula SAE car, in a **10-member team**.
- Sized the shafts by **finite element analysis in SolidWorks** to a **1.2 to 1.5** safety factor on an assumed worst-case shock load.

AWARDS

Research Grant, *Mubadala and Abu Dhabi Maritime Association*

Aug 2024

Industry Recognition and Grant, *Hyundai NGV*

Oct 2022

Full Merit Scholarship, *New York University*

Dec 2021

EXTRACURRICULAR ACTIVITIES

NYU Abu Dhabi Athletics, *Varsity Jiu-Jitsu, Recreational Sport and Team Videographer*

2022 - 2026

- Competed in **jiu-jitsu at university varsity level**, earning selection in each of **four years**.
- Filmed and photographed the university cricket team across a full season, delivering footage and stills each week.
- Recreational sport across padel, golf, tennis, cricket, volleyball, badminton, football, cycling and wall climbing.

Habitat for Humanity Nepal, *Volunteer, Residential Build Programme*

2025

- Constructed a single-storey house from foundation to roof with a volunteer team, handing a finished home to its family.
- Checked dimensions and layouts against the civil drawings on site before the team began work.
- Converted donated labour and materials into permanent housing for families priced out of building.

Emirates Motorsports Organization, *Chief Track Marshal*

2019 - 2025

- Directed trackside safety and first response across **15 events** at Yas Marina Circuit and Dubai Autodrome.

SKILLS

CAD and Design - SolidWorks (CSWA), mechanism design, design for manufacturing, rapid prototyping.

Simulation and Analysis - CFD, finite element analysis (FEA), ANSYS Fluent, OpenFOAM, Converge, Simulink, HPC.

Manufacturing - FDM 3D printing, carbon fibre and fibreglass composites, Arduino, closed-loop PID control.

Programming - Python, MATLAB, machine learning, sensor data acquisition.

Languages - English C2, Urdu C2, Arabic A2.